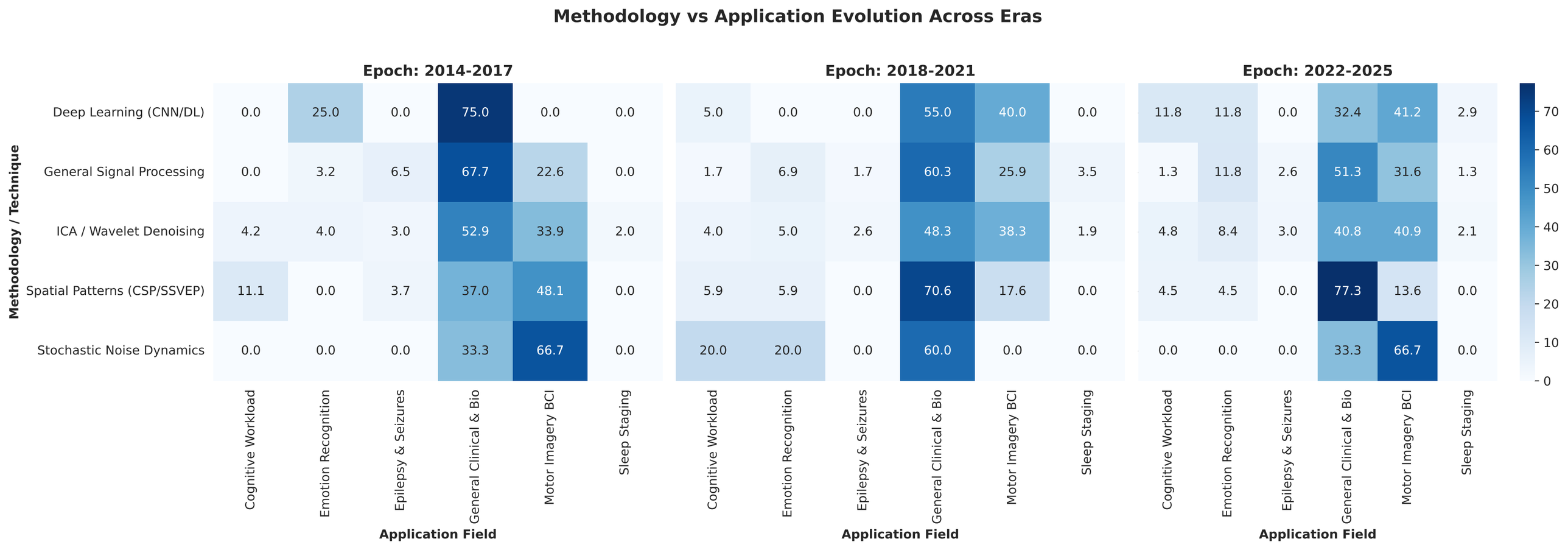
Methodology vs Application Evolution Across Eras
Epoch: 2014-2017
Epoch: 2018-2021
Epoch: 2022-2025
Methodology / Technique
Deep Learning (CNN/DL)
General Signal Processing
ICA / Wavelet Denoising
Spatial Patterns (CSP/SSVEP)
Stochastic Noise Dynamics
0.0 25.0 0.0 75.0 0.0 0.0
0.0 3.2 6.5 67.7 22.6 0.0
4.2 4.0 3.0 52.9 33.9 2.0
11.1 0.0 3.7 37.0 48.1 0.0
0.0 0.0 0.0 33.3 66.7 0.0
5.0 0.0 0.0 55.0 40.0 0.0
1.7 6.9 1.7 60.3 25.9 3.5
4.0 5.0 2.6 48.3 38.3 1.9
5.9 5.9 0.0 70.6 17.6 0.0
20.0 20.0 0.0 60.0 0.0 0.0
11.8 11.8 0.0 32.4 41.2 2.9
1.3 11.8 2.6 51.3 31.6 1.3
4.8 8.4 3.0 40.8 40.9 2.1
4.5 4.5 0.0 77.3 13.6 0.0
0.0 0.0 0.0 33.3 66.7 0.0
Cognitive Workload
Emotion Recognition
Epilepsy & Seizures
General Clinical & Bio
Motor Imagery BCI
Sleep Staging
Application Field
0
10
20
30
40
50
60
70